Nexora Care Capacity Planning Final Report

Power BI methodology, exploratory analysis, forecasting review and recommendations

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Scope of this report
This report documents the methodology and results contained in the submitted Nexora Care Power BI file. It also records project limitations.

Executive summary

Nexora Care requires a consistent view of admissions, discharges, emergency inflow, bed capacity, elective activity and staffing so planners can anticipate pressure and act before capacity becomes constrained. The Power BI solution consolidates six static extracts covering 39 hospitals and presents three report pages: Patient Flow and Admission Overview, Bed Capacity and Staffing Performance, and Capacity Forecast and Scenario Planning.

The historical data contains 1,247,384 distinct admissions and 1,242,074 discharged admissions, leaving a net flow of 5,310 admissions. Elective admissions account for 50.73% of admissions and emergency admissions 49.27%. Out of 1,956,925 distinct ED arrivals, 510,557 resulted in admission, an ED conversion rate of 26.09%. Bed occupancy averages 74.66% of staffed capacity across the recorded observations, but pressure varies materially by weekday: Friday reaches 83.08%, compared with 65.17% on Sunday.

Staffing is consistently below plan. The dataset records 2,040,996 actual staffing assignments against 2,273,506 planned assignments, a gap of 232,510 or approximately 10.23%. The safe-staffing flag is met in 83.99% of staffing records. General Medicine has both the longest average stay, at 5.76 days, and the highest recorded ward occupancy peak, at 136 beds, making it the clearest operational priority in the historical analysis.

Business context and problem

Nexora Care operates across 39 hospitals. Capacity decisions depend on information held in separate admissions, emergency, elective surgery, bed and staffing extracts. When these datasets are reviewed independently, planners cannot easily see whether incoming demand, delayed discharges, scheduled procedures and staffing availability are converging on the same pressure point.

The business problem is to replace fragmented, retrospective monitoring with a central planning view that explains historical demand and gives management earlier visibility of likely bed and staffing pressure. The solution is intended to reduce avoidable elective cancellations and last-minute capacity reallocations. But because the current build uses static files, it can only be described as periodically refreshed decision support rather than a real-time operational system.

Project objectives

- Assess and improve the quality of the six source tables through Power Query.
- Analyse admissions, discharges, ED inflow, occupancy, staffing and elective activity.
- Forecast admissions and occupancy at useful daily and weekly levels.
- Combine known scheduled-elective demand with a modelled emergency component.
- Test flu, emergency-spike and discharge-delay scenarios.
- Provide an interactive decision-support dashboard and actionable recommendations.

Scope boundaries

The analysis covers static data extracts and uses Power Query, the Power BI semantic model and DAX. True hourly forecasting is excluded because bed occupancy is captured in periodic snapshots

rather than at a genuine hourly grain. The data does not support causal linking of each ED visit or elective booking to a specific admission. Automated alerts, production database connections, clinical recommendations, financial forecasting and automated model retraining are outside scope.

Data sources and coverage

Dataset	Rows loaded	Date coverage	Purpose
Admissions and discharges	1,247,498	1 Sep 2024 to 31 Aug 2026	Patient flow, admission type, specialty, discharge and length of stay
Bed occupancy	2,388,750	1 Sep 2024 to 31 Aug 2026	Total, staffed, occupied and closed beds by hospital, ward and bed type
ED arrivals	1,956,944	1 Sep 2024 to 31 Aug 2026	Emergency arrivals, outcomes, triage, arrival mode and time to admission
Elective surgery schedule	1,220,442	1 Sep 2024 to 29 Nov 2026	Known scheduled demand, status, specialty, required bed type and expected stay
Staffing resource	768,690	1 Sep 2024 to 31 Aug 2026	Planned and actual staffing by role, date, hospital and ward
Hospitals	39	Not date based	Hospital name, city, region and size tier

All six sources contain the same 39 hospital identifiers. Elective bookings extend three months beyond the historical admissions, occupancy, ED and staffing period, which creates an appropriate source for known future elective demand.

Data Cleaning Summary

Dataset	Cleaning performed	Missing-value treatment	Duplicate rule	Dataset
Admissions and discharges	Cleaned text, standardised categories, preserved GP/ICU, created date-only fields and recalculated length of stay	Undischarged records retained as blank	Exact duplicates removed; admission IDs checked	Admissions and discharges
Bed occupancy	Standardised wards/bed types and recalculated closed beds	No missing operational values found	Exact duplicates removed	Bed occupancy
ED arrivals	Standardised outcomes, triage and arrival modes	Time to admission left blank for non-admitted cases	Exact duplicates removed	ED arrivals
Elective surgery	Standardised status, specialty and bed type	Missing procedure group labelled “Not Recorded”	Exact duplicates removed	Elective surgery
Staffing	Standardised ward/role and reviewed missing actual staff	Missing actual staff retained pending confirmation	Removed using hospital, ward, date and staff role	Staffing
Hospitals	Standardised hospital names, cities and regions	No material missing values found	Hospital ID checked for uniqueness	Hospitals

Data quality assessment

Issue	Observed condition	Required action
Missing discharge information	5,310 admissions have no discharge datetime, discharge date or calculated length of stay.	Retained as undischarged.
Missing ED time to admission	1,453,909 ED records have no time-to-admission value.	Calculated admission time only for outcome = Admitted.
Missing actual staffing	11,493 staffing rows have no actual_staff value.	Retained as missing values.
Repeated admission identifiers	1,247,498 rows but 1,247,384 distinct admission IDs; five rows appear fully duplicated after load.	Removed duplicate to get the DISTINCTCOUNT of admissions until resolved.

Methodology

Data ingestion and profiling

The six CSV files were imported separately into Power Query. Column names were promoted from the first row and data types were assigned before analytical calculations were created. Profiling considered row counts, uniqueness, missing values, date coverage, category consistency and logical relationships among bed fields and patient-flow dates.

Cleaning and standardisation

- Text fields were trimmed and cleaned to remove leading, trailing and non-printing characters.
- Category labels were converted to consistent proper case. GP and ICU were corrected after case conversion so the abbreviations remained uppercase.
- Blank admission source, discharge destination and procedure-group values were labelled Not Recorded where the absence represented an unknown category rather than a numeric value.
- Exact duplicate removal was applied to admissions, bed occupancy, ED arrivals and elective surgery. Staffing duplicates were assessed using hospital, ward, date and staff role as the composite business key.
- Date-only columns were derived from admission, discharge, occupancy and ED datetimes to support relationships with one shared calendar.
- Length of stay was recalculated as discharge datetime less admission datetime and converted to total hours. Closed beds were recalculated as total beds less staffed beds.
- Missing operational values were not automatically replaced with zero because zero and unknown have different meanings.

Semantic model

The intended design is a star schema with Hospitals and CalendarTable as shared dimensions and five fact tables: admissions and discharges, bed occupancy, ED arrivals, elective surgery, and staffing resource. The hospital relationships are correctly configured as single-direction many-to-one relationships.

DAX and metric design

Forty-seven measures are present in the model. Distinct counts are used for admissions, ED arrivals and surgeries. Occupancy is calculated as the ratio of total occupied beds to total staffed beds within the current filter context. Average daily staffing measures first total staffing by date and then average those daily totals. Scenario measures add assumed surge and discharge-delay effects to baseline occupied beds. The accompanying DAX library contains each exact formula, purpose, format, visual usage, validation status and recommended correction.

Exploratory analysis

EDA examined time trends, admission type, admissions versus discharges, ED arrivals and admissions, specialty length of stay, weekday occupancy, occupied versus staffed capacity, staffing fulfilment and recorded ward peaks. Interpretation focuses on association and operational pattern.

Forecasting approach

The capacity page contains a native Power BI forecast applied to daily Bed Occupancy Rate. The configuration uses a 30-day forecast horizon, 95% confidence and seasonality of seven days. This is useful as an initial occupancy forecast because it reflects the weekly cycle visible in the historical series.

Scenario simulation

Three What-if parameters are provided: Emergency Spike from 0% to 50% in 5% steps, Flu Surge from 0% to 50% in 5% steps, and Discharge Delay from 0 to 3 days in 0.5-day steps. Scenario Bed Demand adds the three impacts to Average Occupied Beds. Scenario Occupancy Rate compares the result with Average Staffed Beds, while Available Beds, Bed Shortfall and Capacity Status translate the result into planning outputs.

Limitations

- The source is static and does not provide live operational status.
- Occupancy records are periodic snapshots; true hourly forecasting is not supported.

Conclusion

The Nexora Care analysis establishes a useful foundation for capacity planning. It brings six operational extracts into one Power BI solution and identifies defensible patterns in patient flow,

weekday occupancy, specialty length of stay and staffing delivery. The clearest historical risks are Friday occupancy pressure, General Medicine demand and a persistent staffing gap against plan.